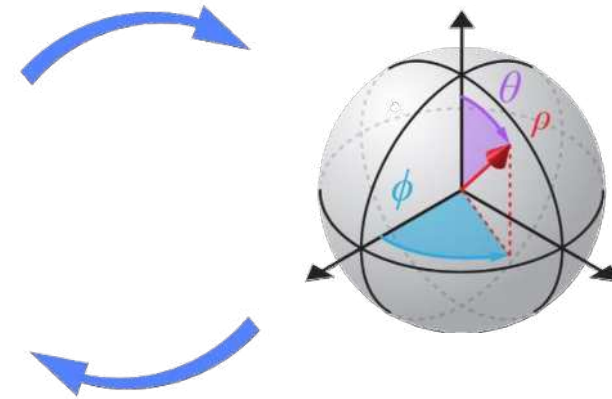
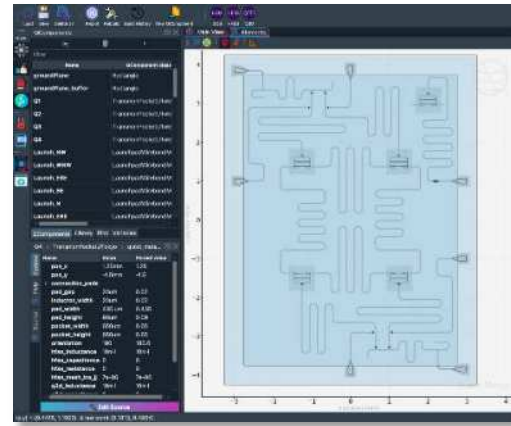




Zlatko K. Minev

IBM Quantum
IBM T.J. Watson Research Center
Yorktown Heights, NY

Overview of Quantum Hardware Design Energy, Circuits, and Metal



@zlatko_minev

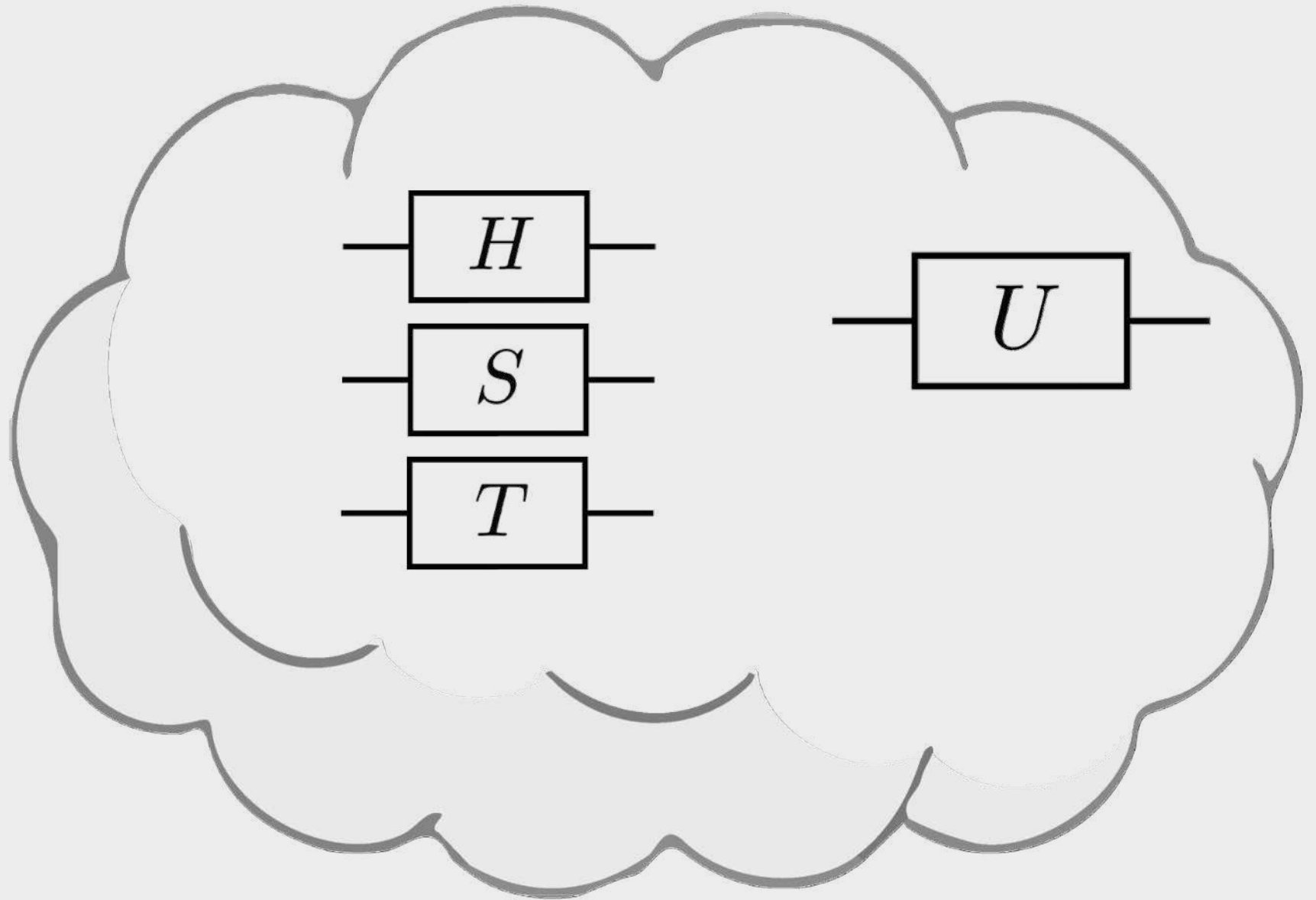


zlatko-minev.com



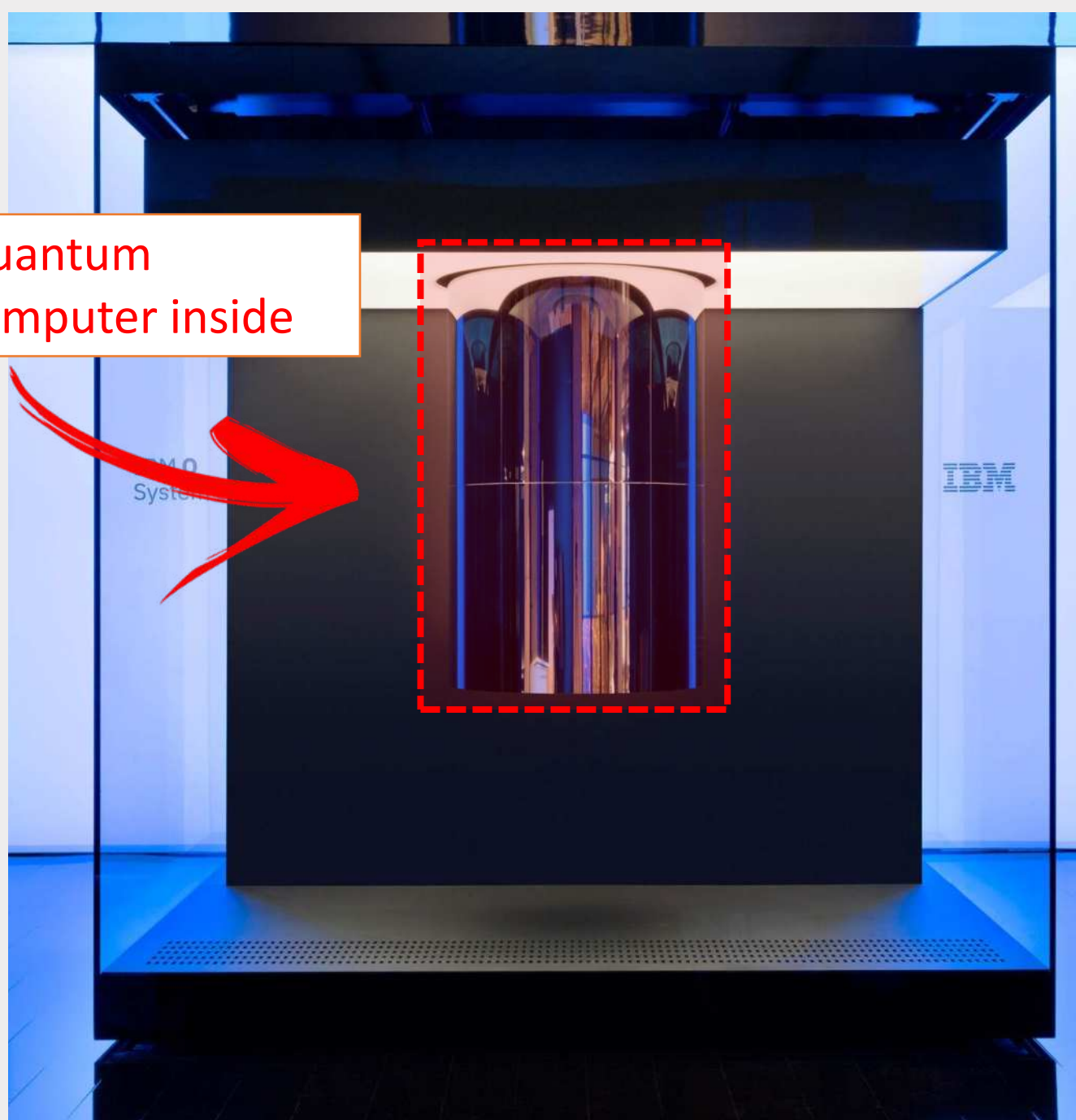
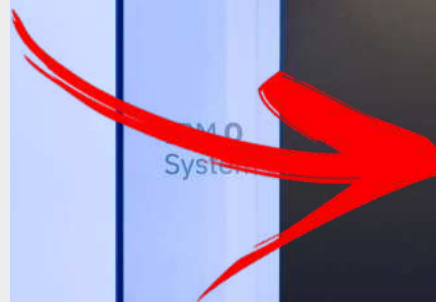
qiskit.org/metal

"Quantum cloud"

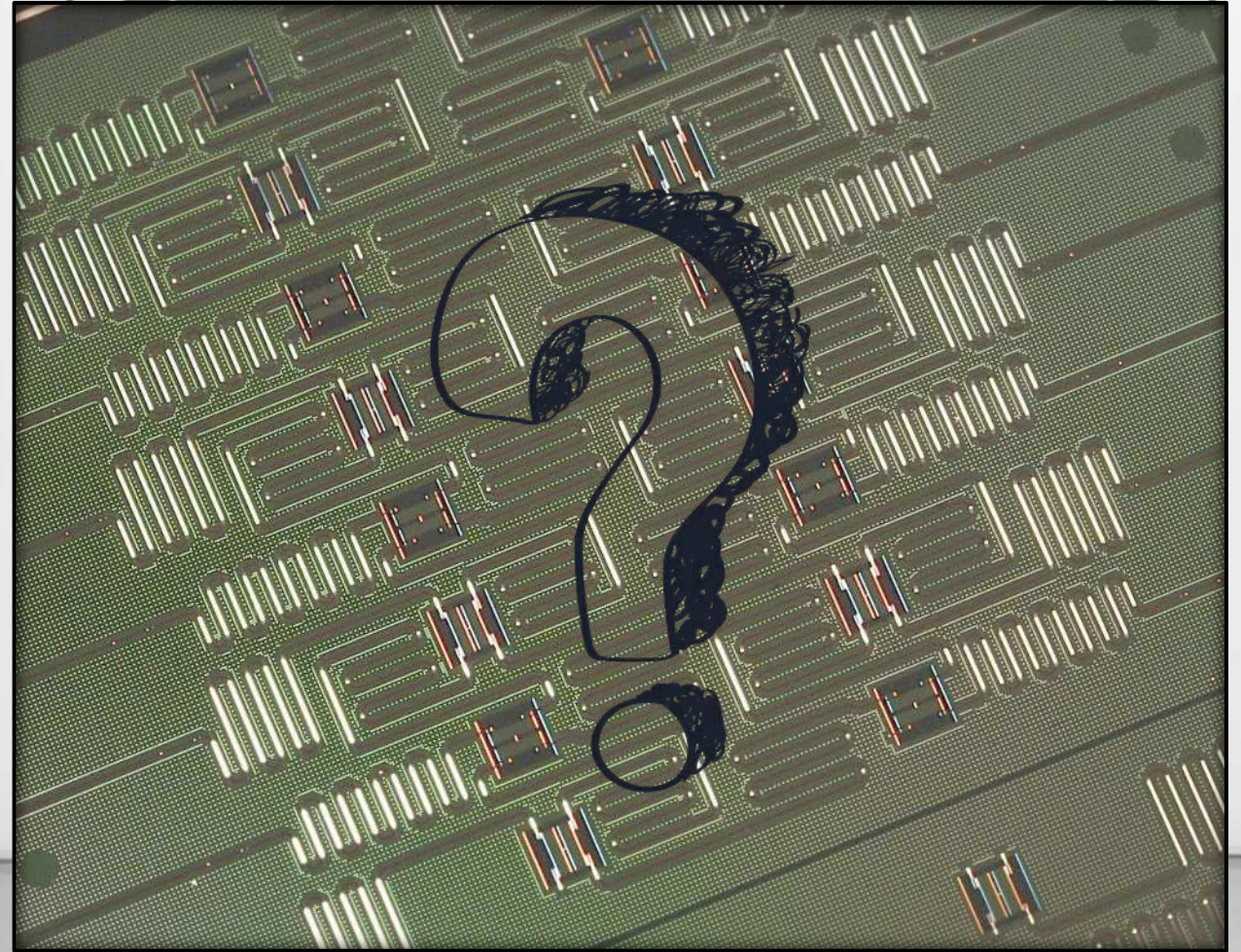


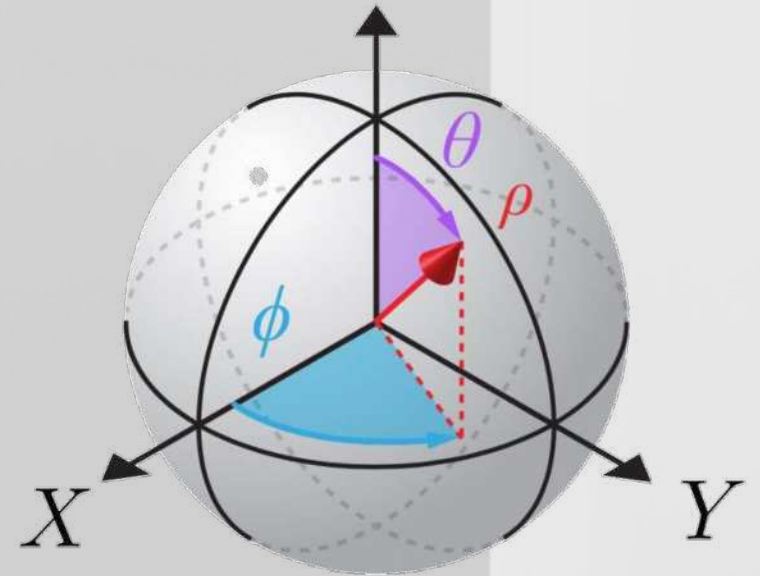
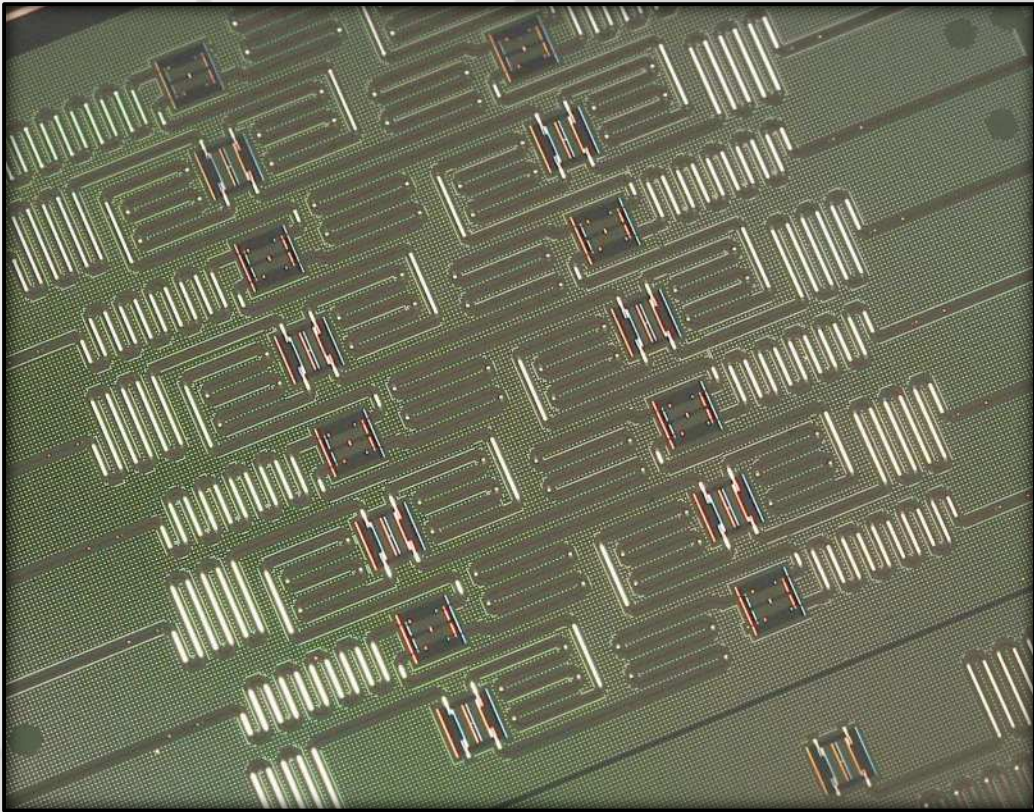


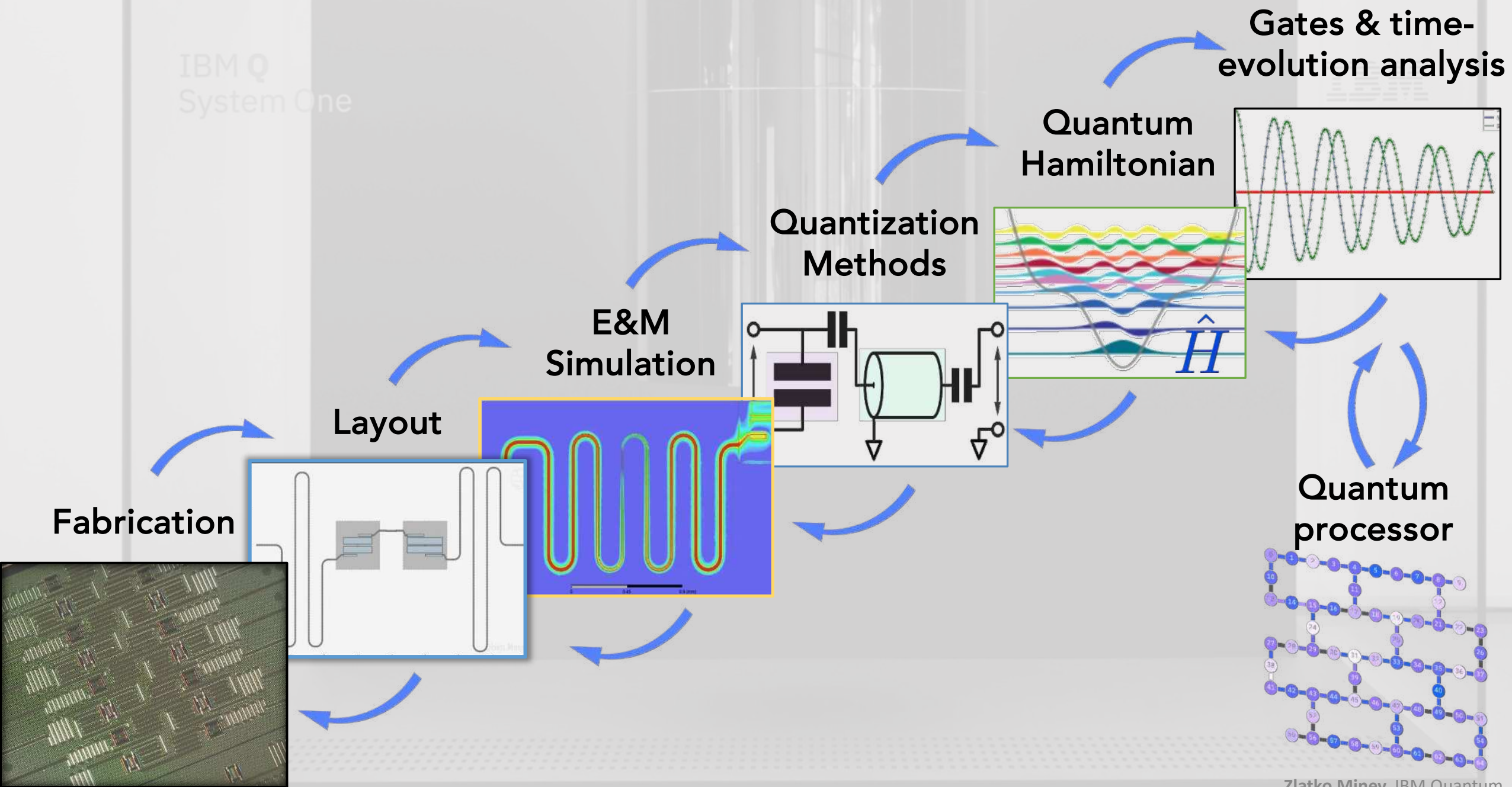
Quantum
computer inside



Quantum
processor



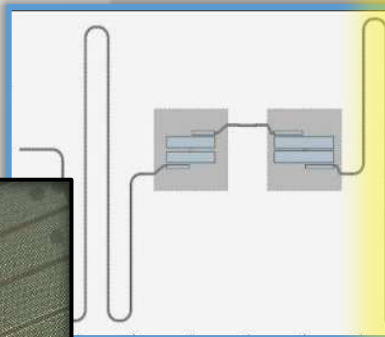
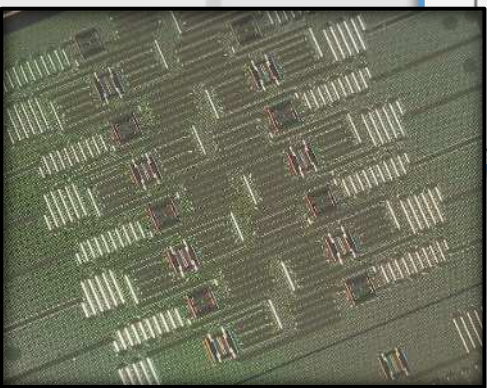




IBM Q
System One

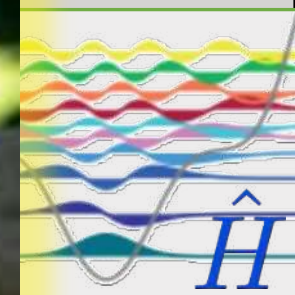
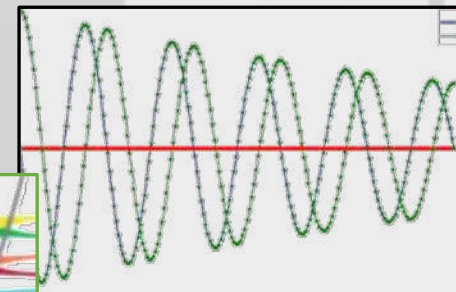
Layout

Fabrication

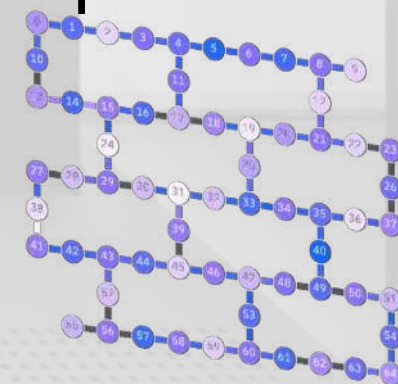


Gates & time-
evolution analysis

quantum
miltonian



Quantum
processor



Layout



E&M Simulation



Quantization Methods



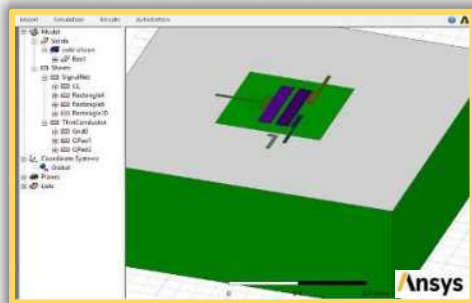
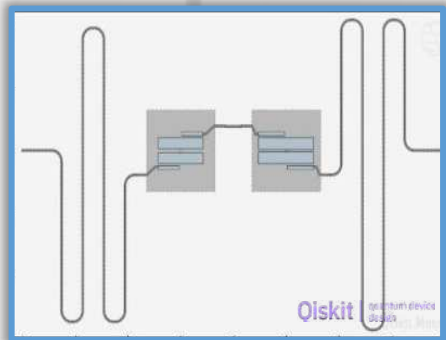
Quantum Hamiltonian



Gates & time-evolution analysis

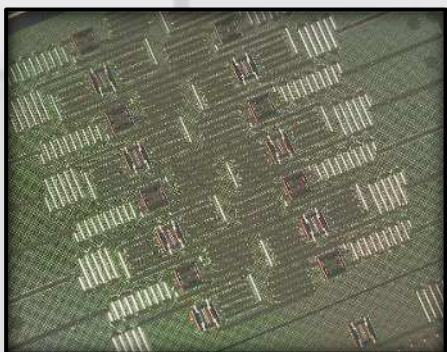
IBM Q
System One

IBM

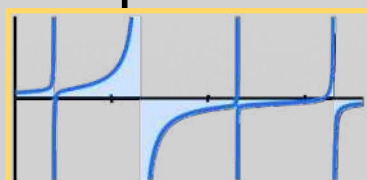


Fabrication

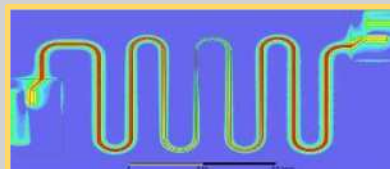
DRC, LVS



impedance



eigenmode



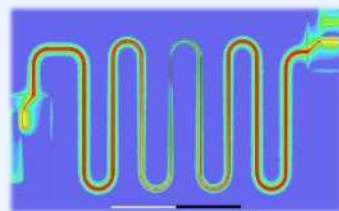
capacitive

$$\begin{pmatrix} n_0 \\ n_1 \\ n_2 \\ n_3 \\ n_4 \\ n_5 \end{pmatrix} \begin{pmatrix} n_0 & n_1 & n_2 & n_3 & n_4 & n_5 \\ C_{0\Sigma} & -C_{01} & -C_{02} & -C_{03} & -C_{04} & -C_{05} \\ & C_{1\Sigma} & -C_{12} & -C_{13} & -C_{14} & -C_{15} \\ & & C_{2\Sigma} & -C_{23} & -C_{24} & -C_{25} \\ & & & C_{3\Sigma} & -C_{34} & -C_{35} \\ & & & & C_{4\Sigma} & -C_{45} \\ & & & & & C_{5\Sigma} \end{pmatrix}$$

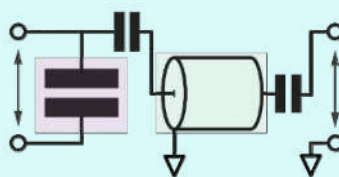
impedance



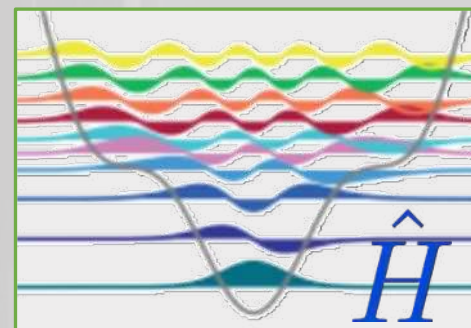
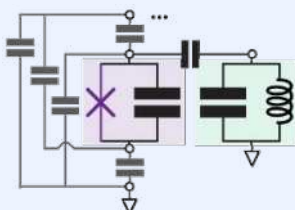
energy



quasi-lumped



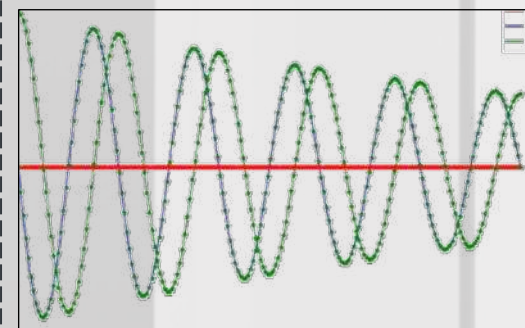
lumped



$$\hat{H} = \hat{H}_{\text{sys}} + \hat{H}_{\text{int}}$$

$$\hat{H}_{\text{full}} = \hat{H}_0 + \sum_{n=1}^K \hat{H}_n + \sum_{n=0}^K \sum_{m=n+1}^K \hat{H}_{nm}$$

MPS
Diagonalization
Dissipation
..



Rotating frames
Time evolution
Driven dynamics
Floquet
Dissipation
...

Layout



E&M Simulation



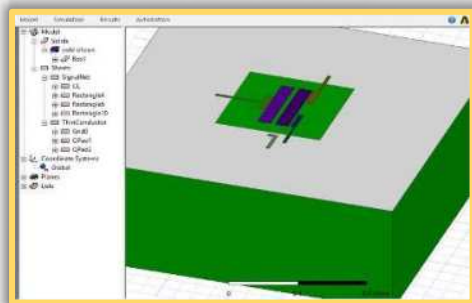
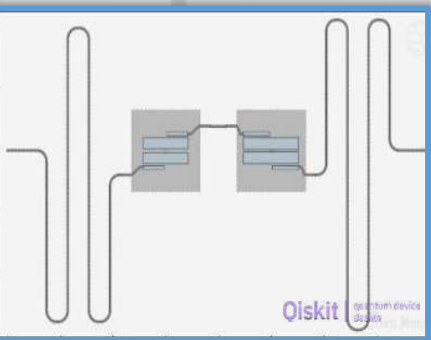
Quantization Methods



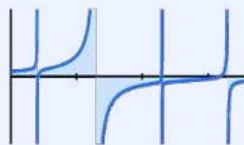
Quantum Hamiltonian



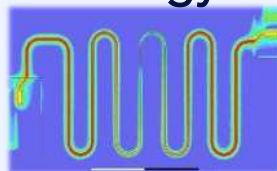
Gates & time-evolution analysis



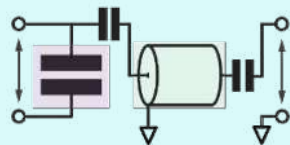
impedance



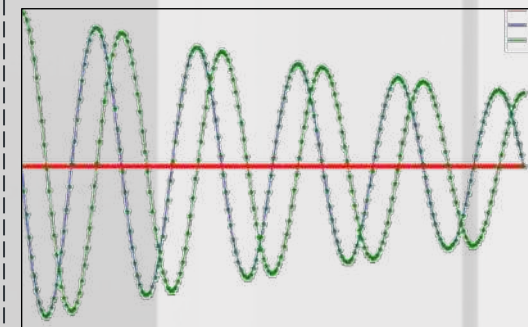
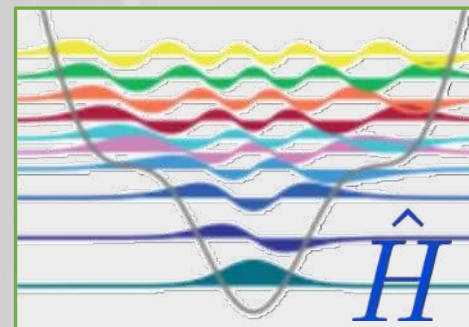
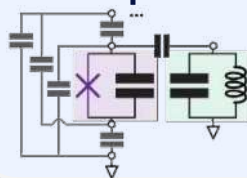
energy



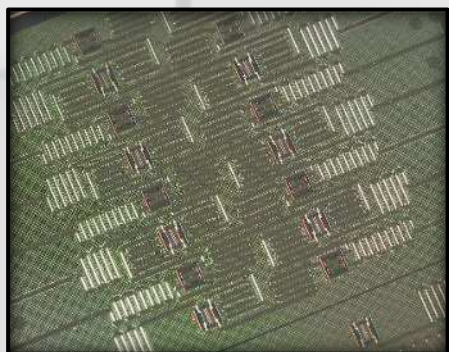
quasi-lumped



lumped

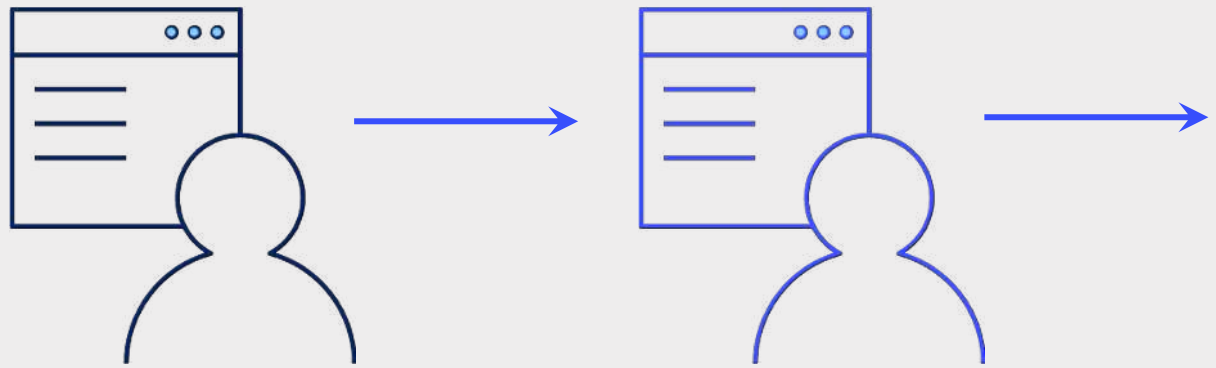


Fabrication



Quantum Device Design

Make it easy!



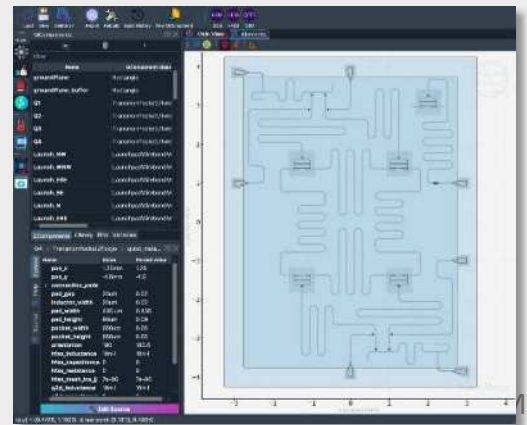
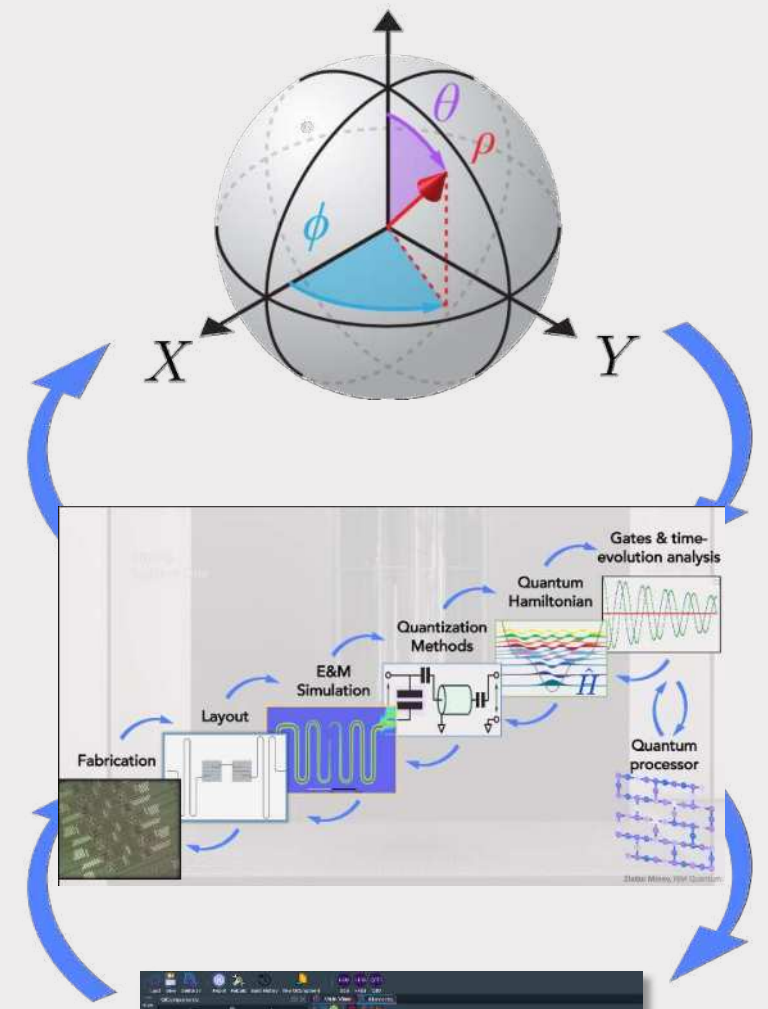
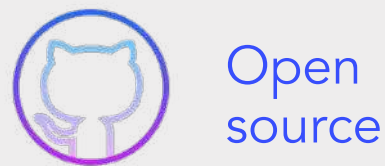
Classical
developer /
engineer

Enabling innovation
of quantum computer
hardware & analysis

Quantum

Engineering

Software



Quantum

A few introductory reviews

And many more... check online or ask us for specific topic

Mineev, Z. Lec. 16-22, *Introduction to Quantum Computing and Quantum Hardware* (2020) qiskit.org/learn/intro-qc-qh

Blais, A., Grimsmo, A. L., Girvin, S. M., & Wallraff, A. (2020) *Circuit Quantum Electrodynamics* (arXiv:2005.12667)

Kjaergaard, M., Schwartz, ... Oliver, W. D. (2020) *Superconducting Qubits: Current State of Play*
Annual Reviews of Condensed Matter Physics 11, 369-395

Krantz, P., Kjaergaard, M., Yan, F., ... & Oliver, W. D. (2019) A quantum engineer's guide to superconducting qubits
Applied Physics Reviews, 6(2), 021318

Corcoles, A. D., Kandala, A., ... Gambetta, J. M. (2019) Challenges and Opportunities of Near-Term Quantum Computing Systems. *Proceedings of the IEEE*, 1–15.

Wendin, G. (2017) Quantum information processing with superconducting circuits. *RPP*, 80(10), 106001

Gambetta, J. M., Chow, J. M., & Steffen, M. (2017) Building logical qubits in a superconducting quantum computing system. *Npj Quantum Information*, 3(1), 2

Girvin, S. M. (2011) Circuit QED: superconducting qubits coupled to microwave photons. *Quantum machines: measurement and control of engineered quantum systems*, 113, 2.

Clerk, A. A., Girvin, S. M., Marquardt, F., & Schoelkopf, R. J. (2010) Introduction to quantum noise, measurement, and amplification
Reviews of Modern Physics, 82(2), 1155–1208

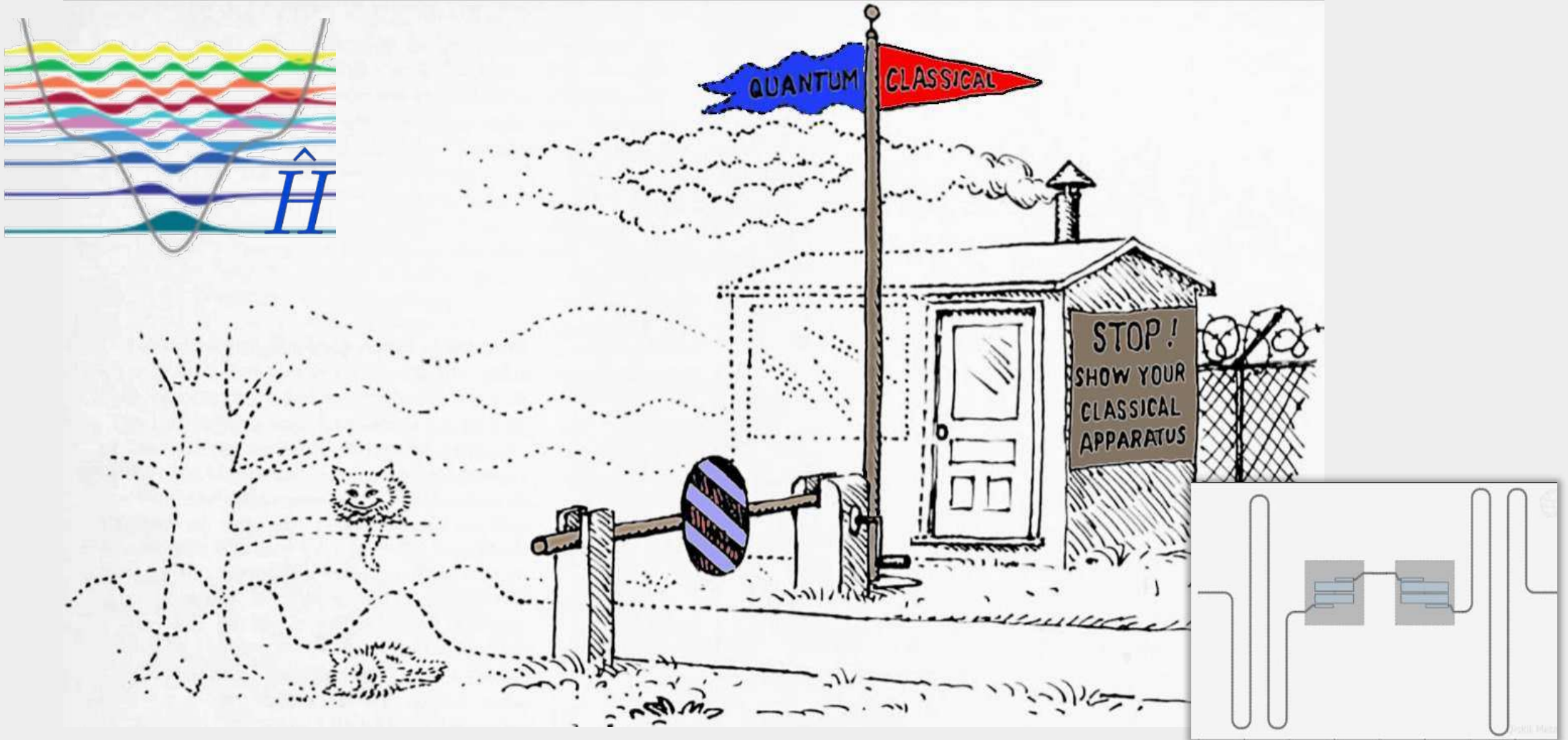
Clarke, J., & Wilhelm, F. K. (2008) Superconducting quantum bits. *Nature*, 453(7198), 1031–1042

Devoret, M. H. (1997) Quantum Fluctuations in Electrical Circuits.
In *Fluctuations Quantiques/Quantum Fluctuations* (p. 351)

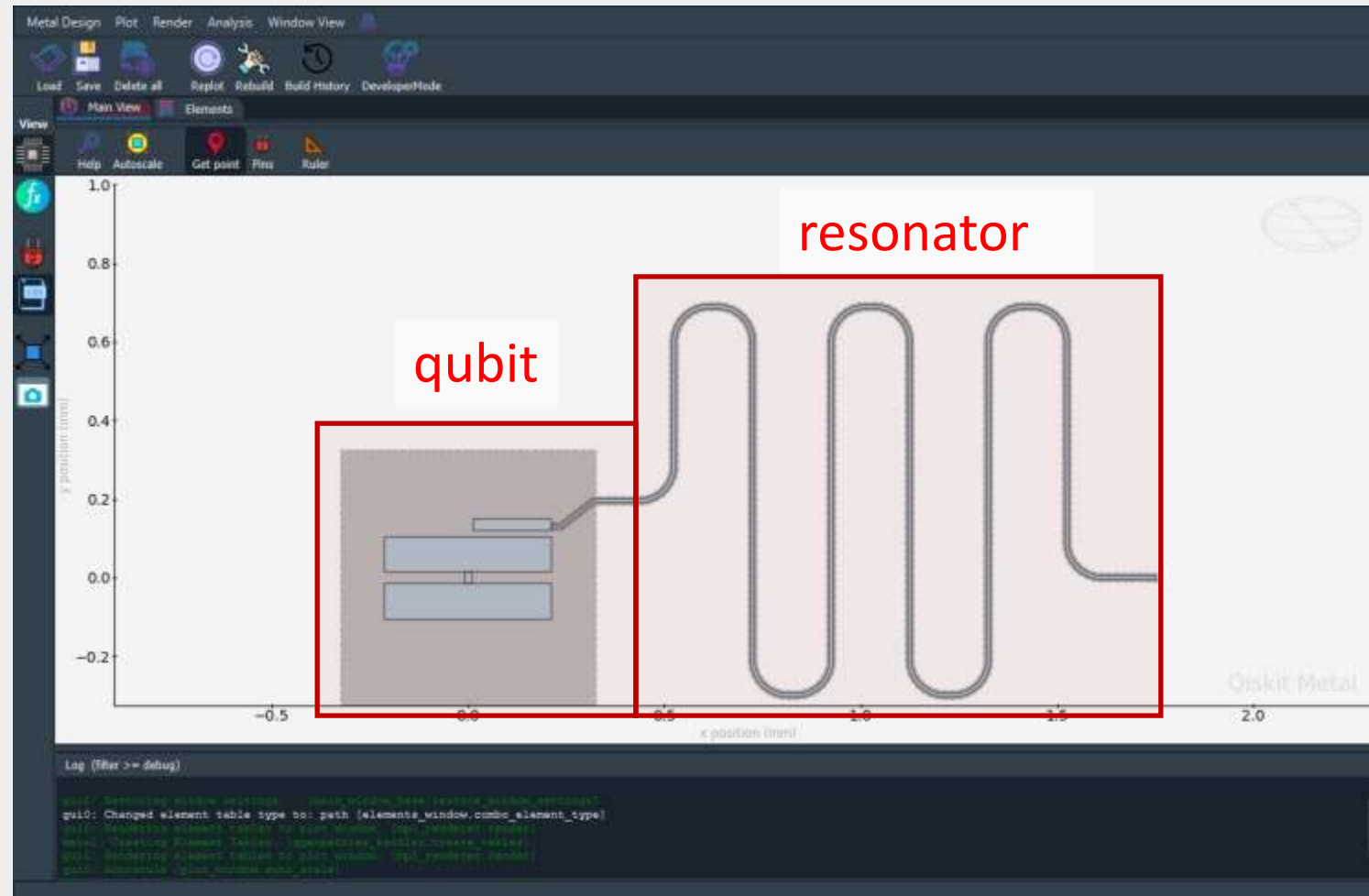
...

Quantum Analysis

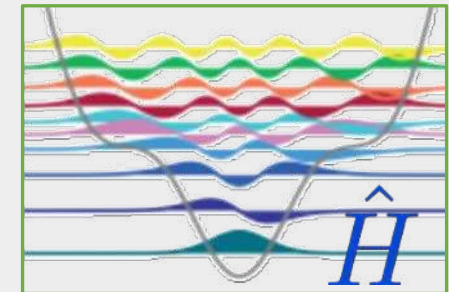
Quantum Analysis



Simple example

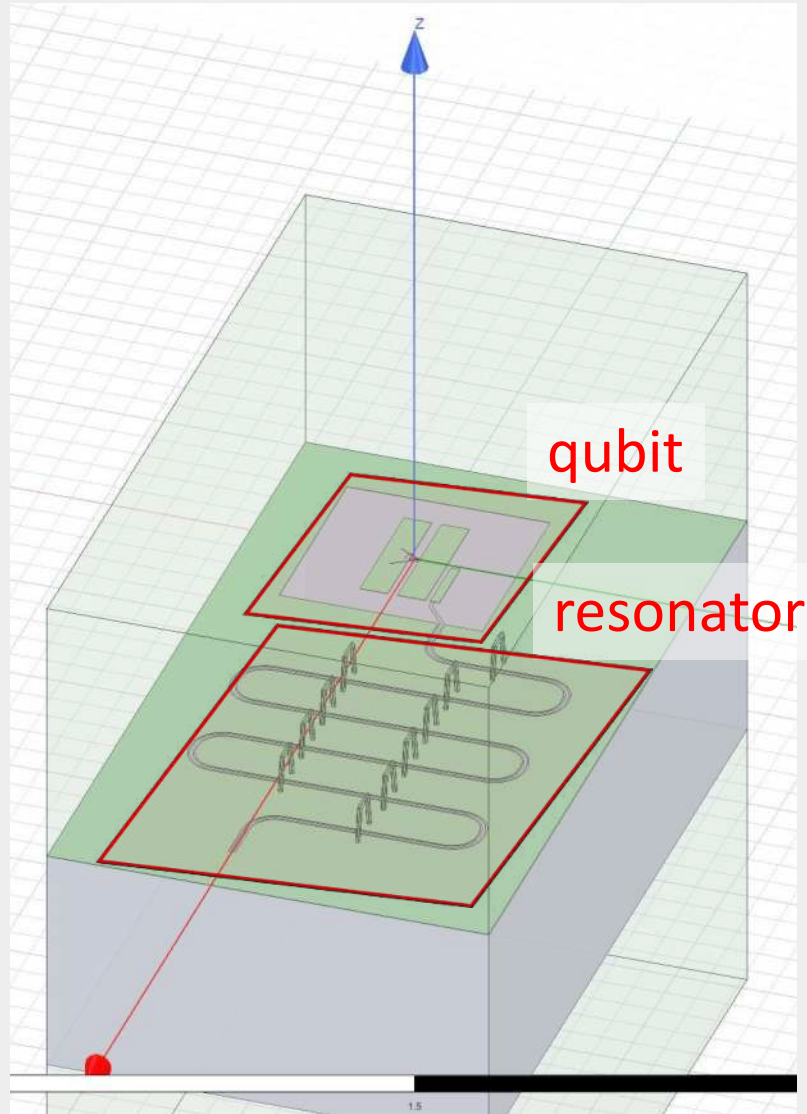


?

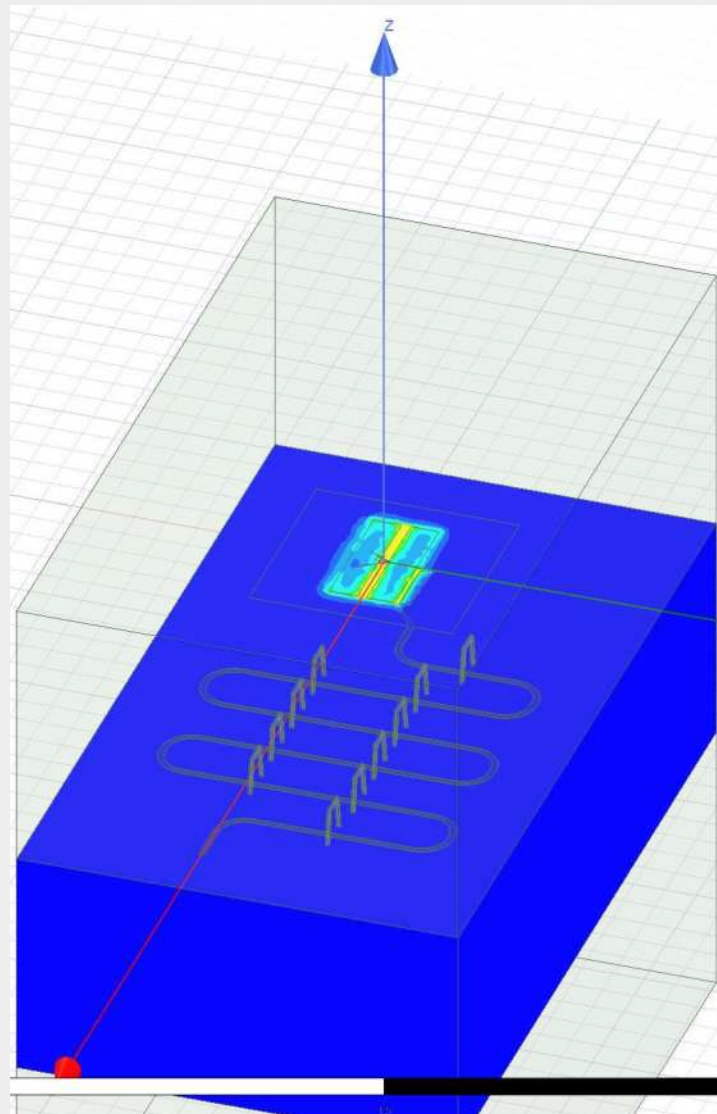


Modeling step 1: Classical E&M approximations

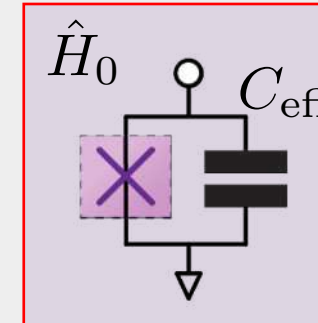
Physical (linear) model



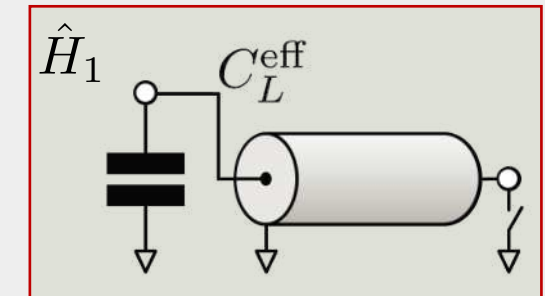
Eigenmodes



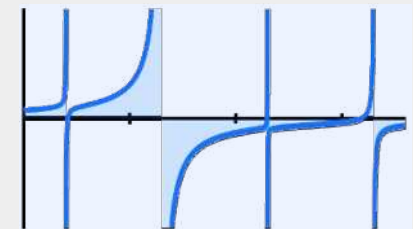
Quasi lumped models



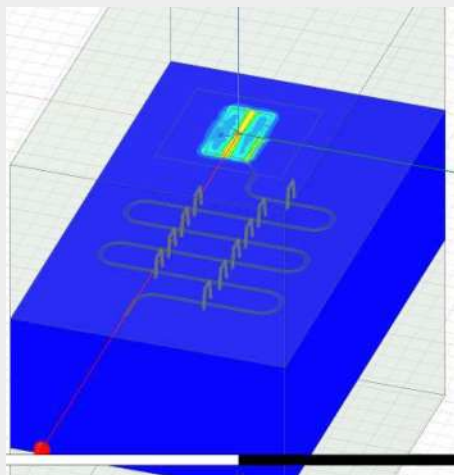
$$\begin{matrix} n_0 & n_1 & n_2 & n_3 & n_4 & n_5 \\ \begin{matrix} n_0 \\ n_1 \\ n_2 \\ n_3 \\ n_4 \\ n_5 \end{matrix} & \begin{pmatrix} C_{0\Sigma} & -C_{01} & -C_{02} & -C_{03} & -C_{04} & -C_{05} \\ & C_{1\Sigma} & -C_{12} & -C_{13} & -C_{14} & -C_{15} \\ & & C_{2\Sigma} & -C_{23} & -C_{24} & -C_{25} \\ & & & C_{3\Sigma} & -C_{34} & -C_{35} \\ & & & & C_{4\Sigma} & -C_{45} \\ & & & & & C_{5\Sigma} \end{pmatrix} \end{matrix}$$



Impedance



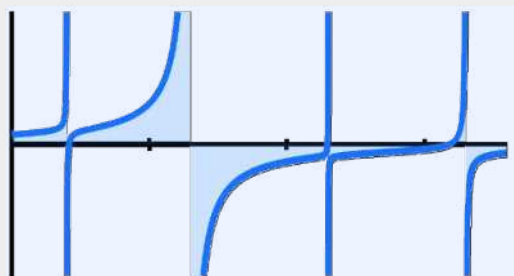
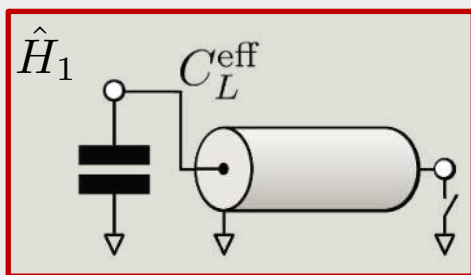
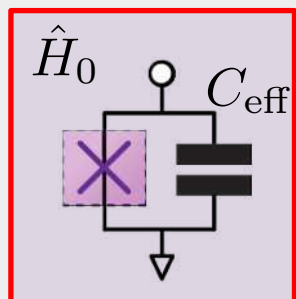
Modeling step 2: Quantize



$$\hat{H}_{\text{tot}}$$

$$\hat{H}_{\text{tot}} = \hat{H}_{\text{sys}} + \hat{H}_{\text{int}}$$

$$\hat{H}_{\text{tot}} = \hat{H}_{\text{lin}} + \hat{H}_{\text{nl}}$$



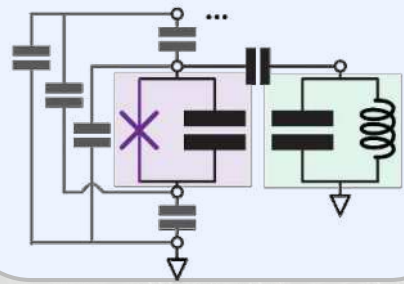
There's a description for every job!

Step 2: Landscape of quantization methods

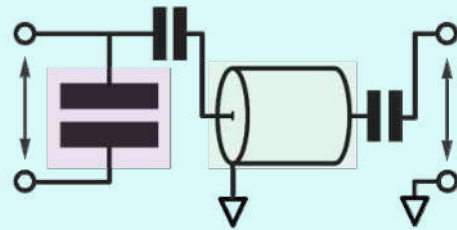
quasi-static

full-wave

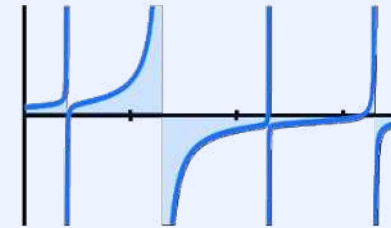
lumped



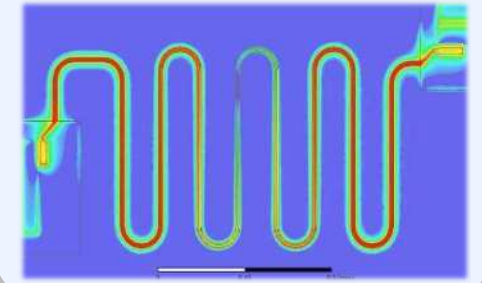
quasi-lumped



impedance



energy



fast

more information, complexity, accuracy

scQubits

QuCAT

pyEPR

...

See Friday Workshop!

QuTiP

KQC

Qiskit | quantum device design
Metal

Yurke & Denker (1984), Devoret (1997), Burkard *et al.* (2004), Koch *et al.*...

Malekakhlagh *et al.* (2017, 2019), Gely *et al.* (2019), Parra-Rodriguez *et al.* (2019), Minev *et al.* (2021), ...

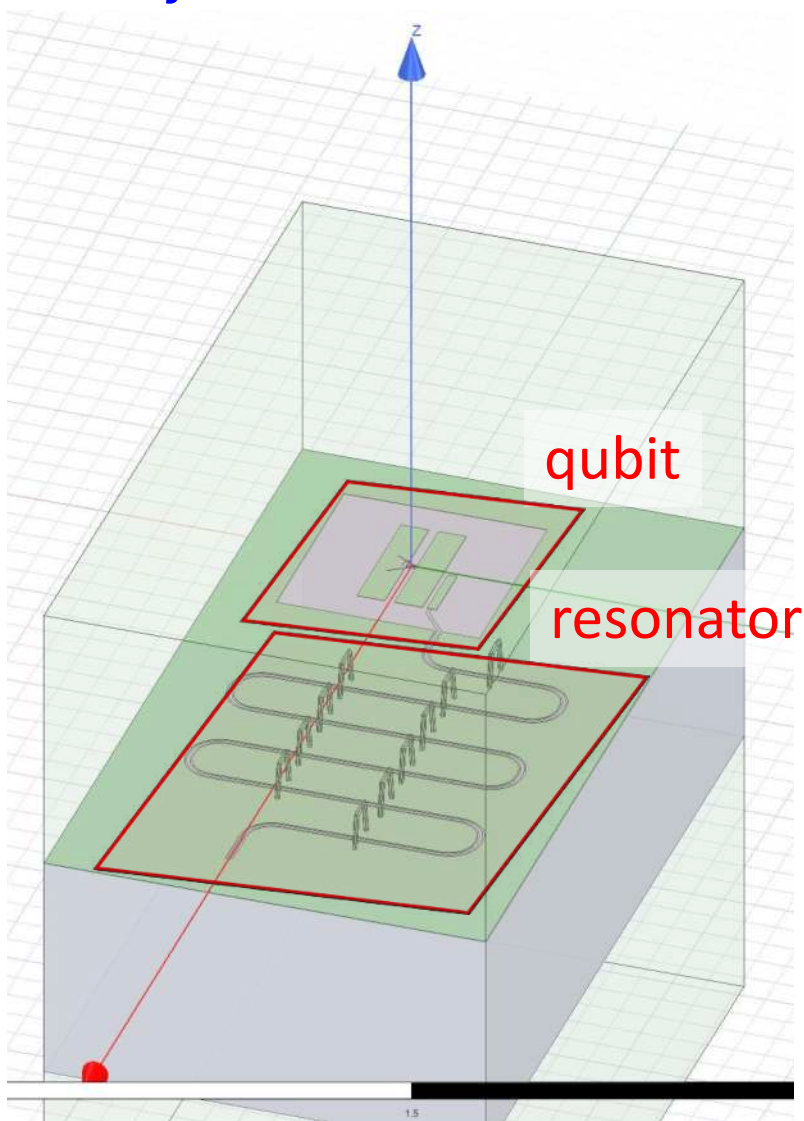
Nigg *et al.* (2012), Bourassa *et al.* (2012), Solgun *et al.* (2014, 2015, 2017) ...

Minev (2018)
Minev *et al.* (2020)

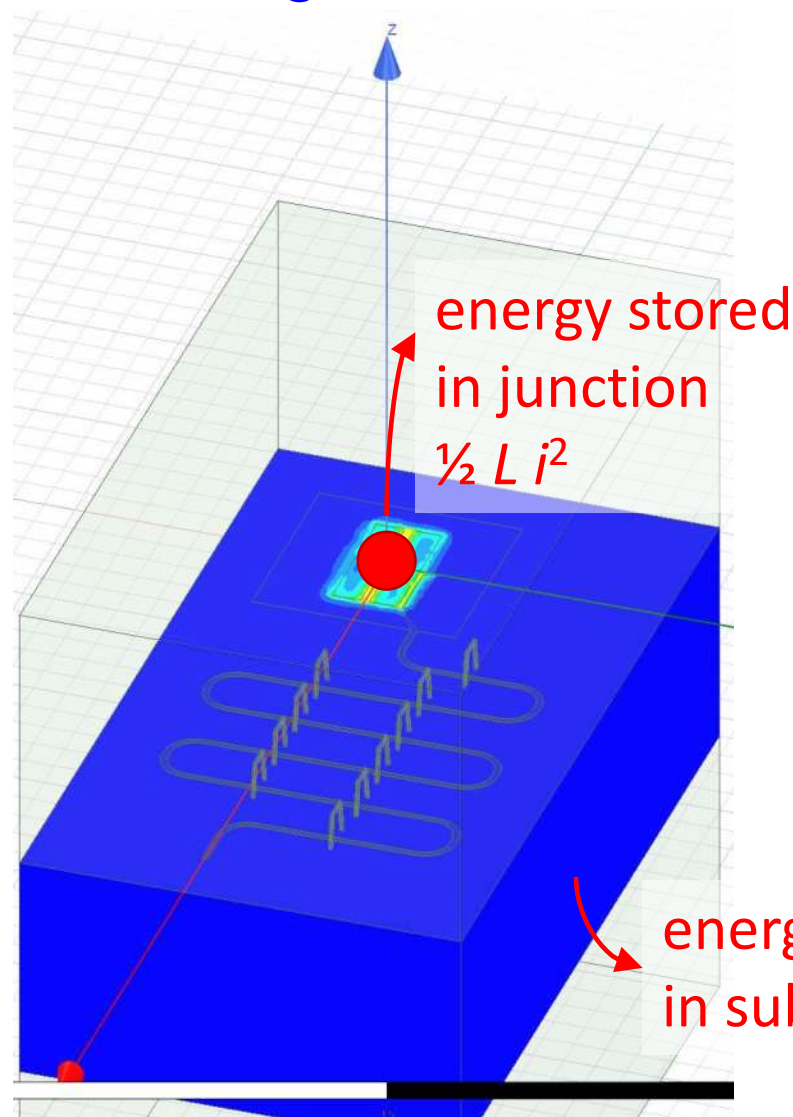
Zlatko Minev, IBM Quantum (17)

Example of step 2 with EPR

Physical (linear) model



Eigenmodes



Non-linear device
params

$$p_j \longrightarrow \hat{H}_{\text{full}}$$

Dissipation
budget

$$p_l \longrightarrow \mathcal{D}[\sqrt{\kappa} \hat{a}]$$

Step 3: Hamiltonian

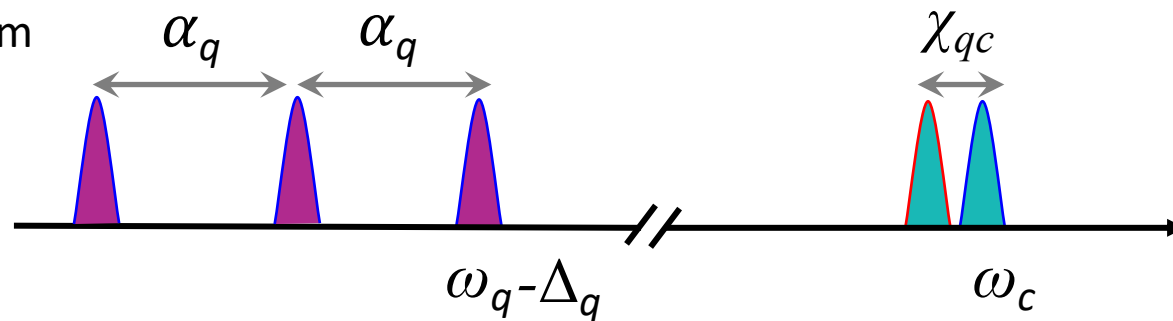
Qubit/cavity anharmonicity

$$\alpha_{q/c} = p_{q/c}^2 \frac{\hbar \omega_{q/c}^2}{8E_J}$$

Qubit-cavity dispersive shift

$$\chi_{qc} = p_q p_c \frac{\hbar \omega_q \omega_c}{4E_J}$$

Transition spectrum

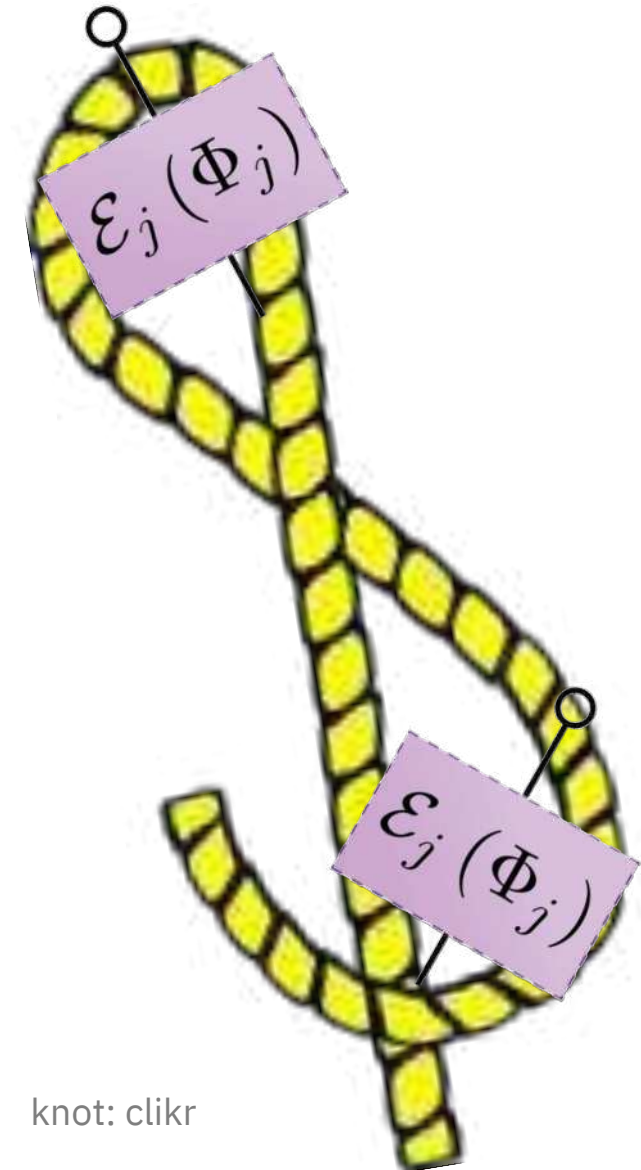
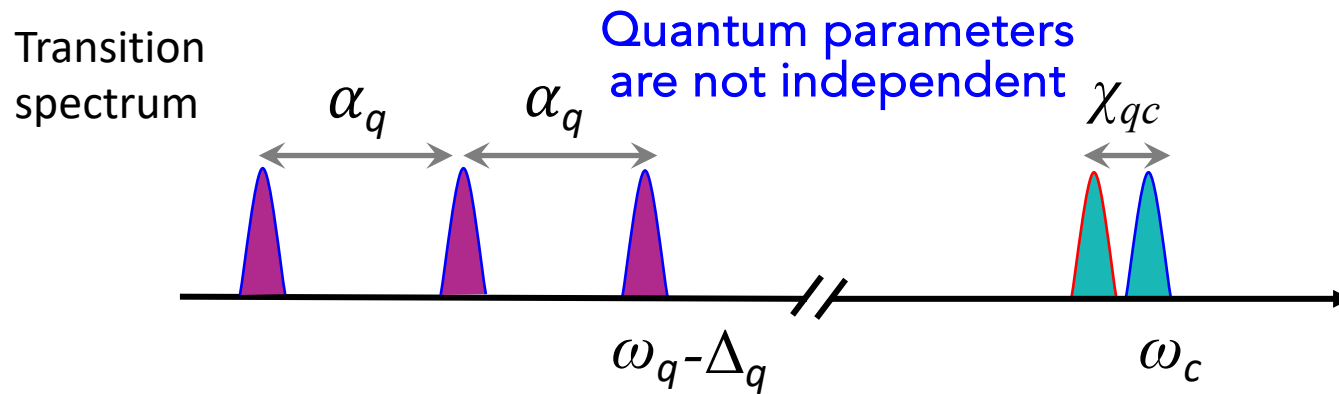
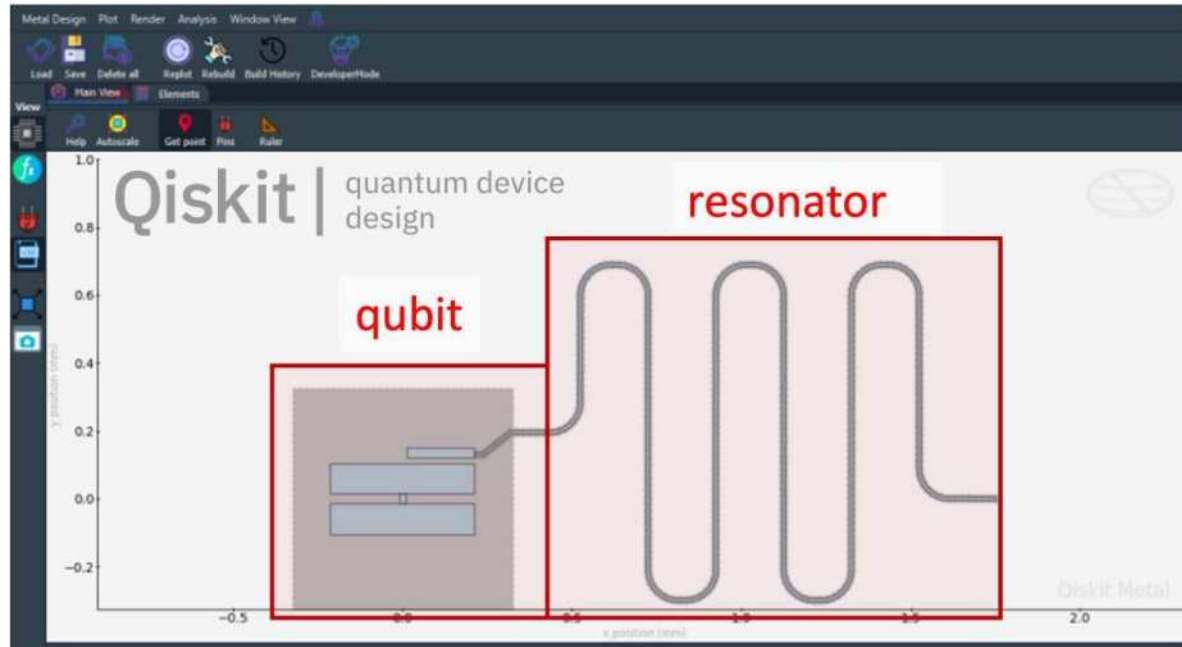


Qubit Lamb shift

$$\Delta_q = \alpha_q - \frac{1}{2}\chi_{qc}$$

H_{eff} : for simplicity, showing up to $\mathcal{O}(\varphi^6)$ in RWA

Step 3: Hamiltonian: Universal constraints & EPR monogamy



knot: clikr

H_{eff} : for simplicity, showing up to $\mathcal{O}(\varphi^6)$ in RWA

EPR theory vs. experiment

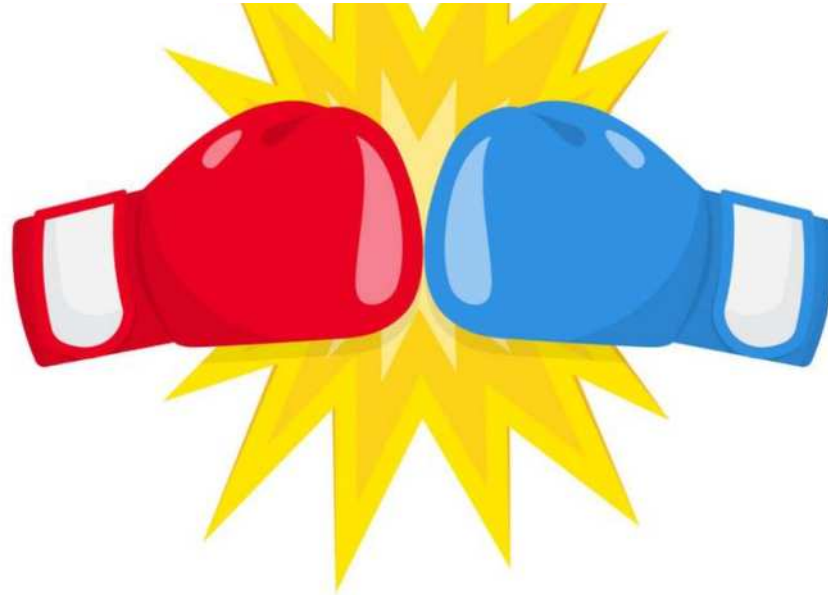
arXiv.org > quant-ph > arXiv:2010.00620

Quantum Physics

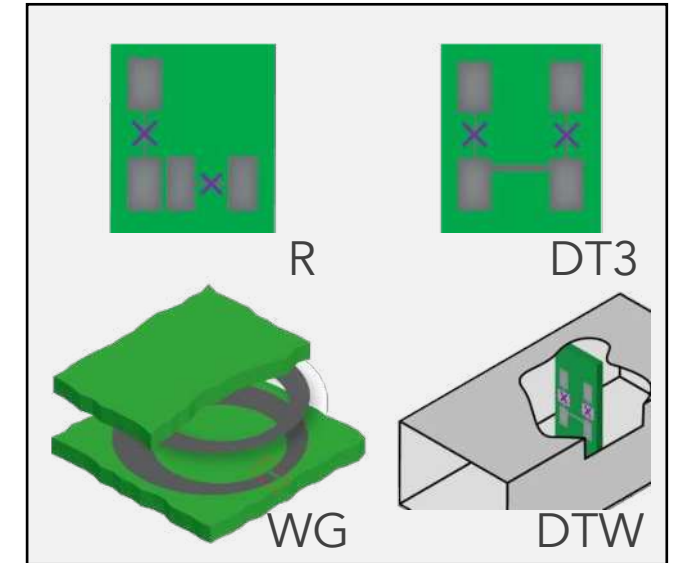
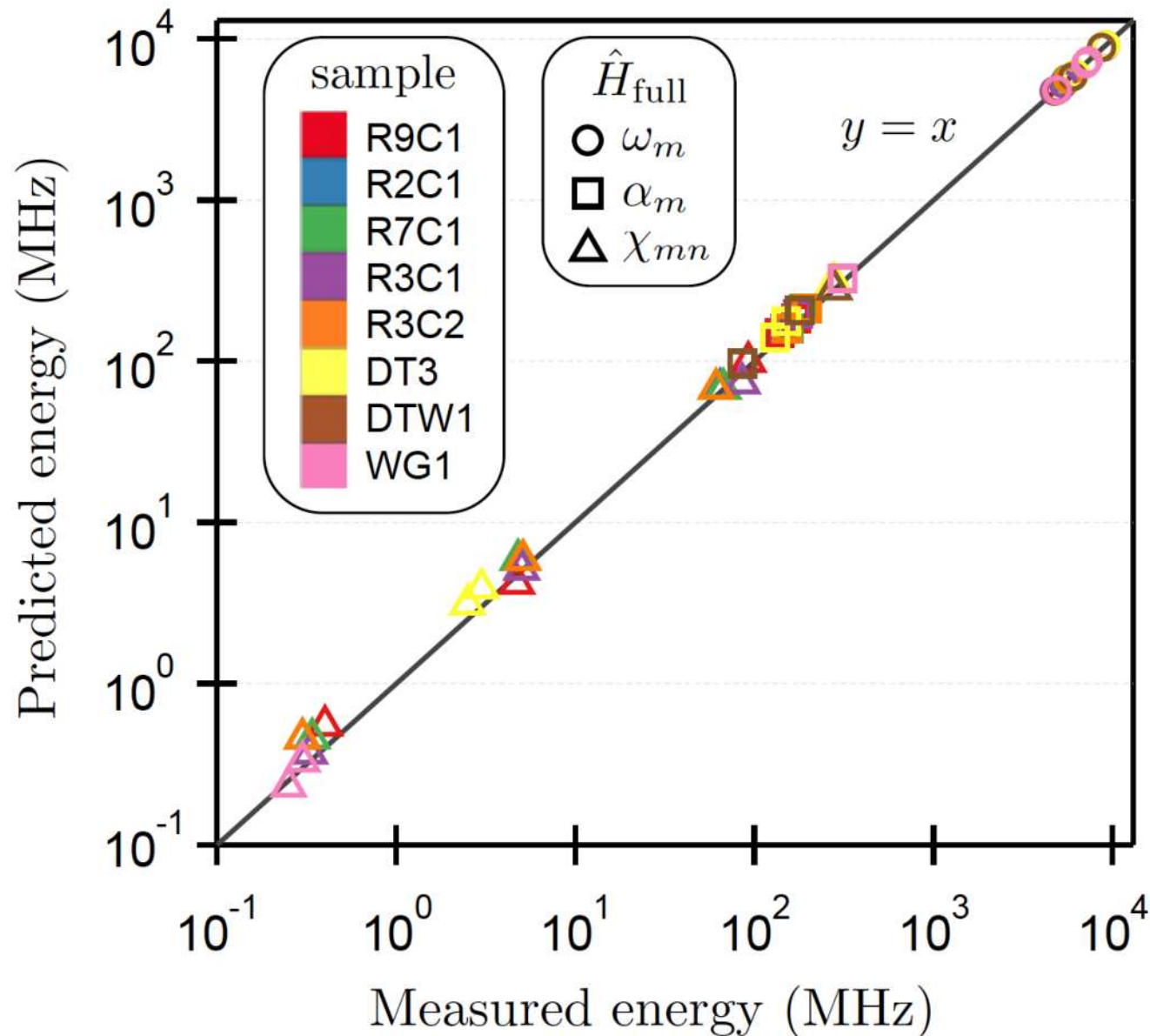
[Submitted on 1 Oct 2020]

Energy-participation quantization of Josephson circuits

Zlatko K. Mineev, Zaki Leghtas, Shantanu O. Mundhada, Lysander Christakis, Ioan M. Pop, Michel H. Devoret



Theory vs. experiment: agreement over 5 orders of magnitude



R: Minev *et al.* (2018)

WG: Minev *et al.* (2013, 2016)

DT3, DTW: Minev *et al.* (2019)

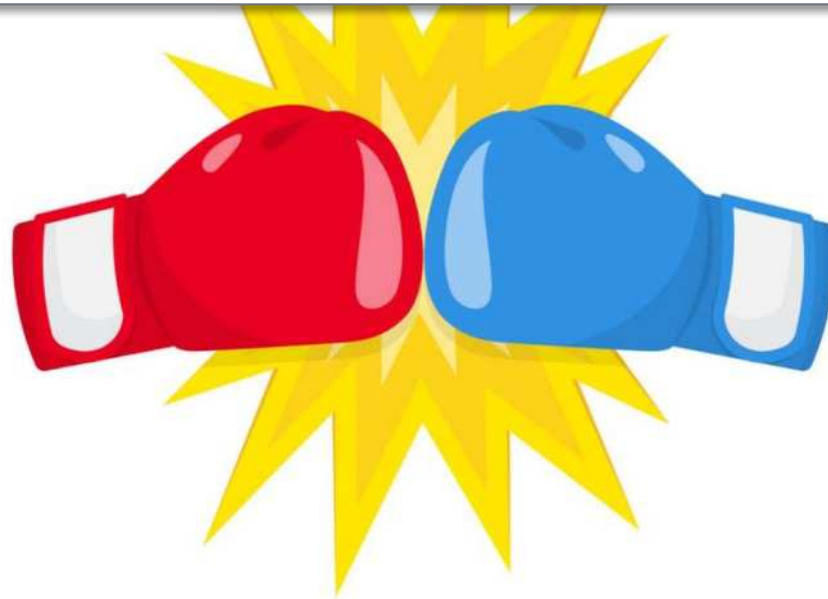
Planar devices & comparison to other methods

Quantum Physics

[Submitted on 2 Feb 2021]

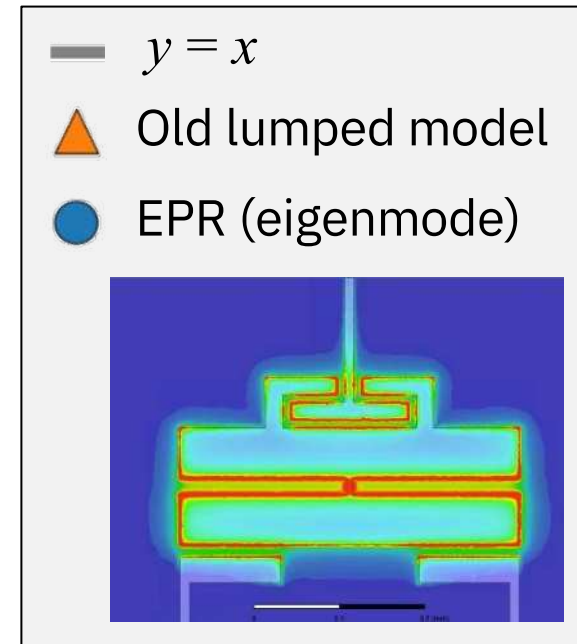
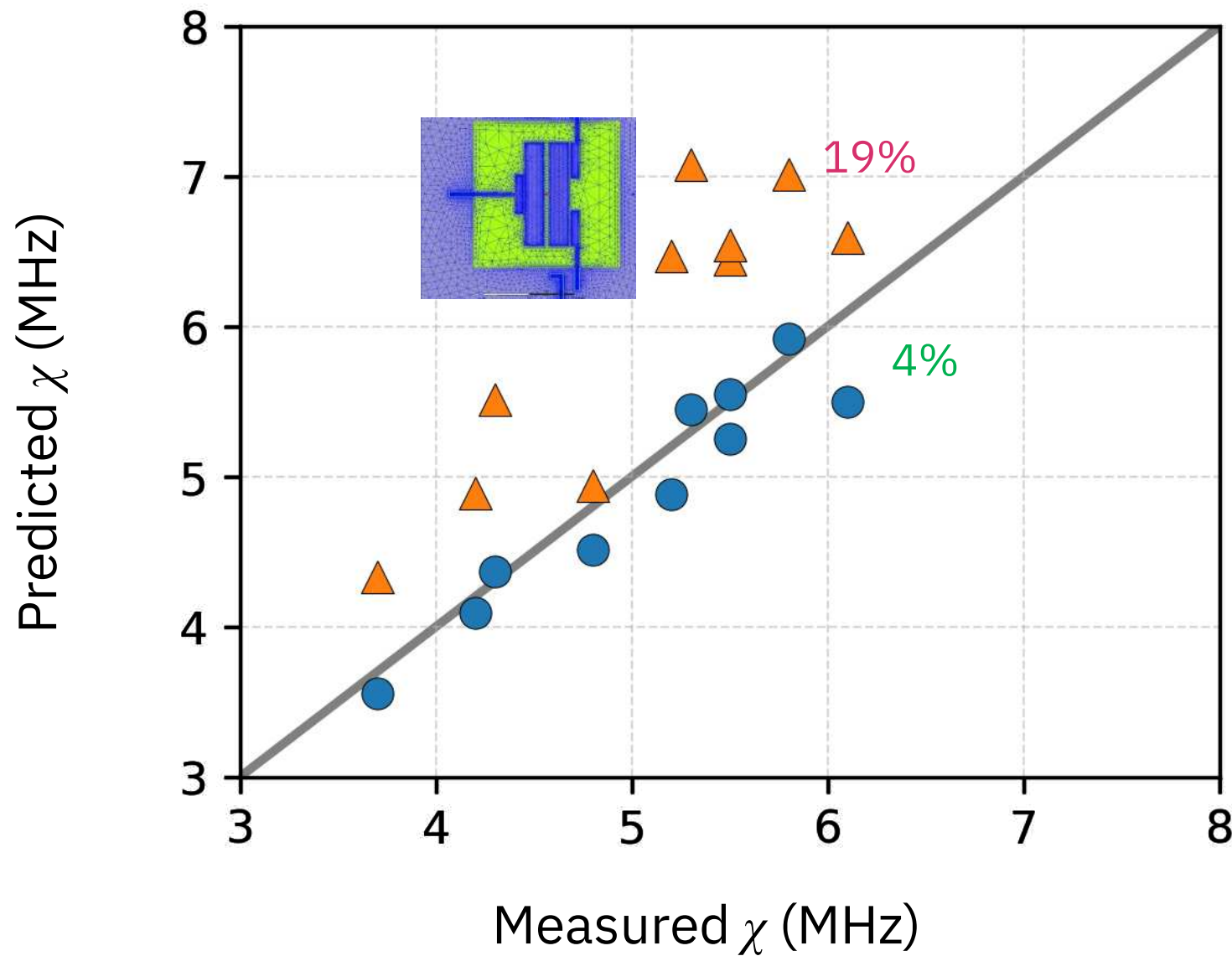
Exploiting dynamic quantum circuits in a quantum algorithm with superconducting qubits

Antonio D. Corcoles, Maika Takita, Ken Inoue, Scott Lekuch, Zlatko K. Minev, Jerry M. Chow, Jay M. Gambetta



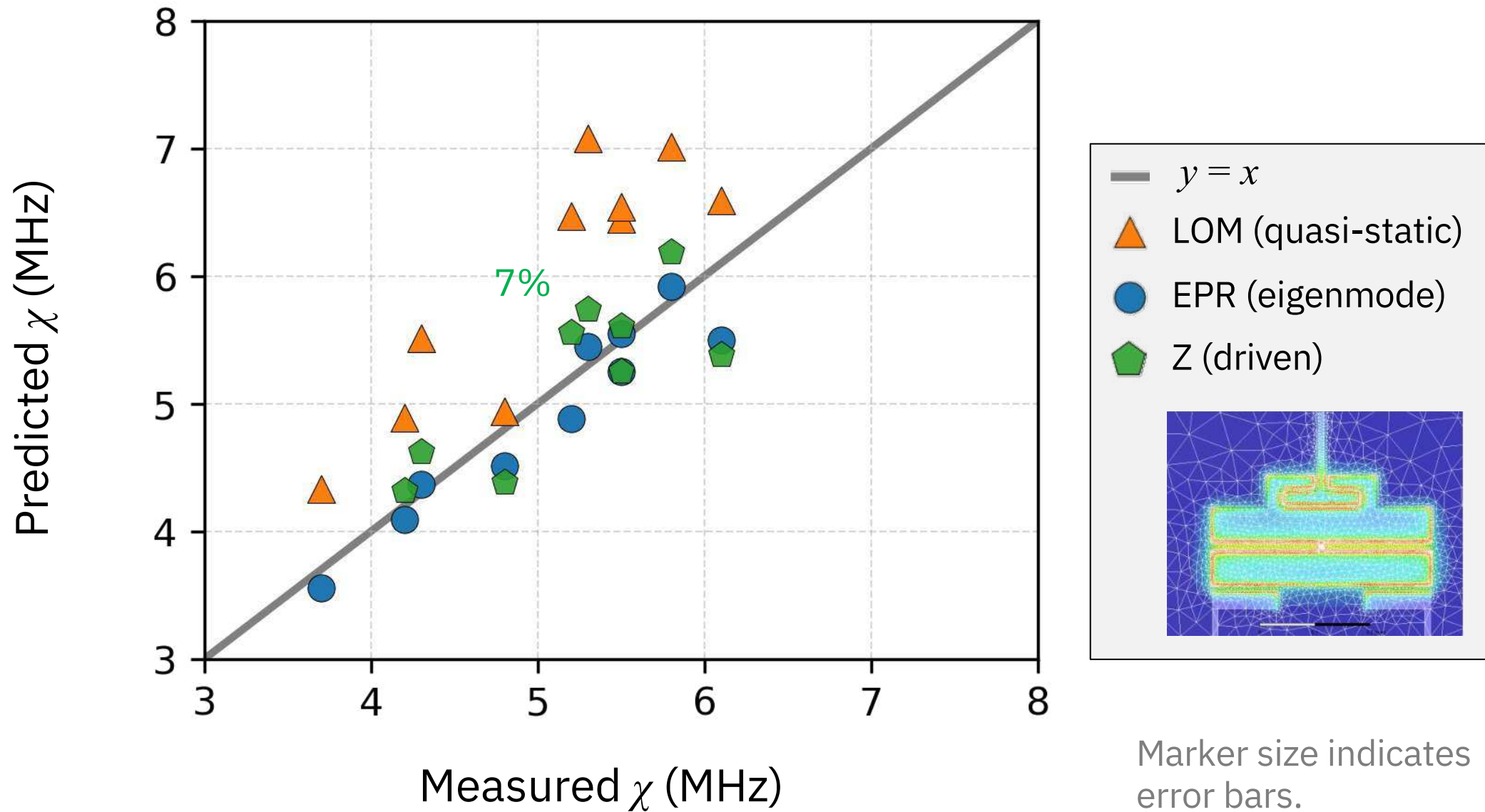
Automated with
Qiskit | quantum device
design

Measured vs. predicted: qubit-readout cross-Kerr



Marker size indicates error bars.

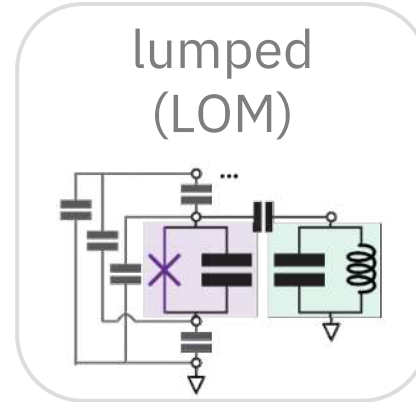
Measured vs. predicted: qubit-readout cross-Kerr



Quantization approaches



Quasi-static



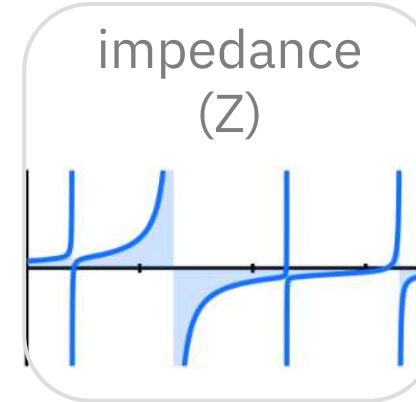
19% / 10%

higher

fastest and
cheapest

low

Full-wave

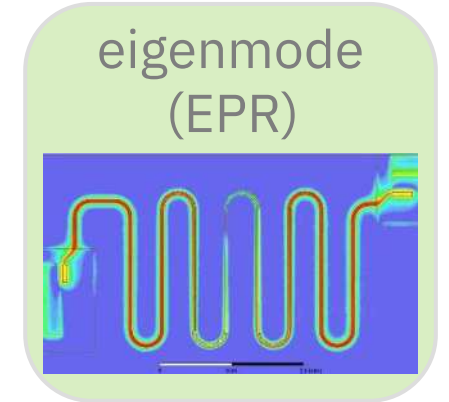


7%

low

moderate

high



4%

low

moderate

highest*

pyEPR 🍺!

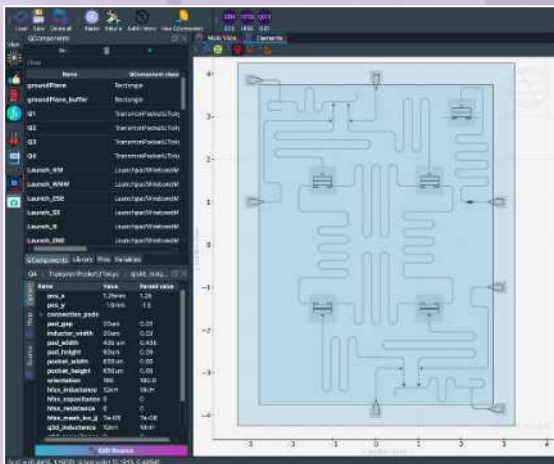
Agreement
for presented results

External inputs

Speed

Generality

* includes dissipative params



+ pyEPR 🍺!

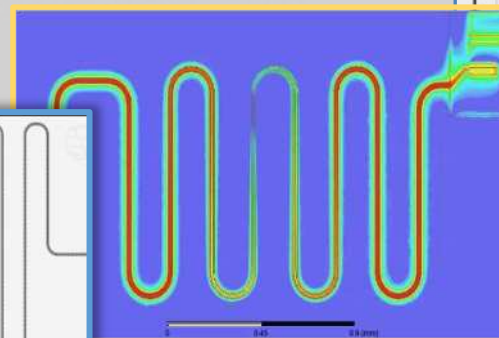
+ SCQubits ☒

+ QuTiP

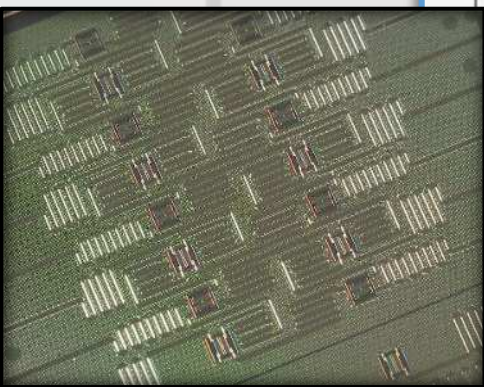
+ ⋮

E&M
Simulation

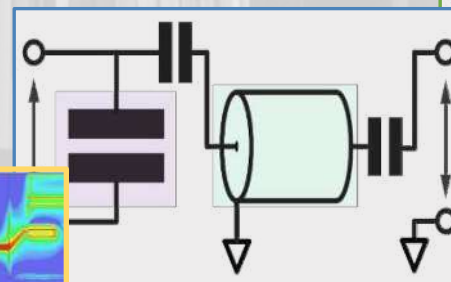
Layout



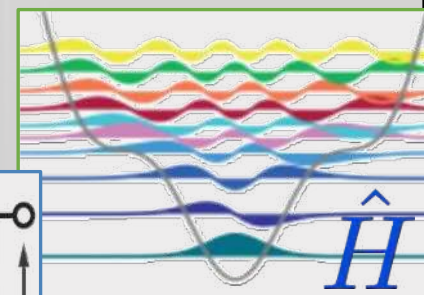
Fabrication



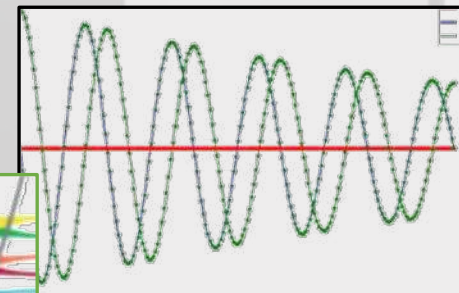
Quantization
Methods



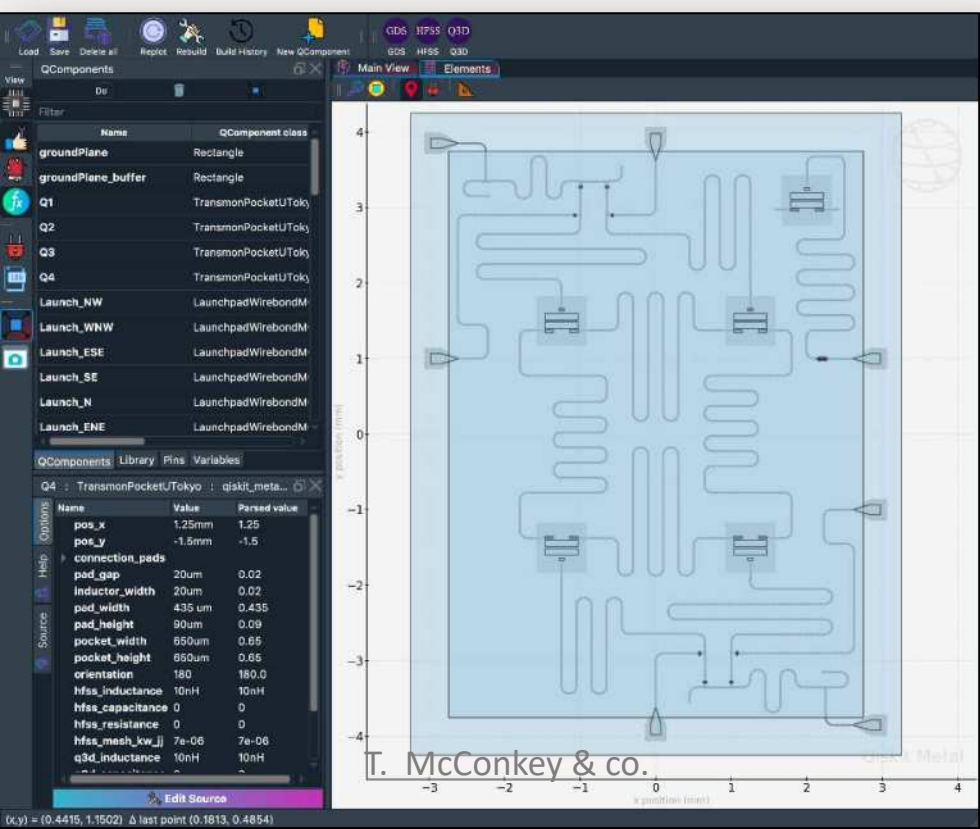
Quantum
Hamiltonian



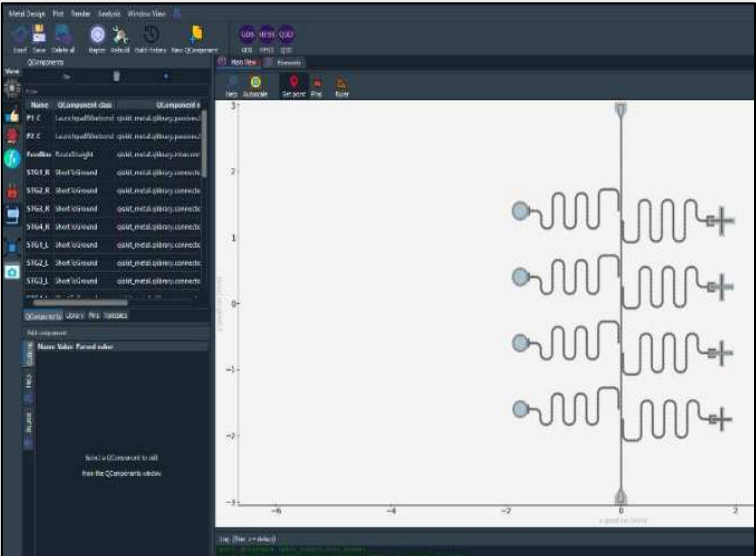
Gates & time-
evolution analysis



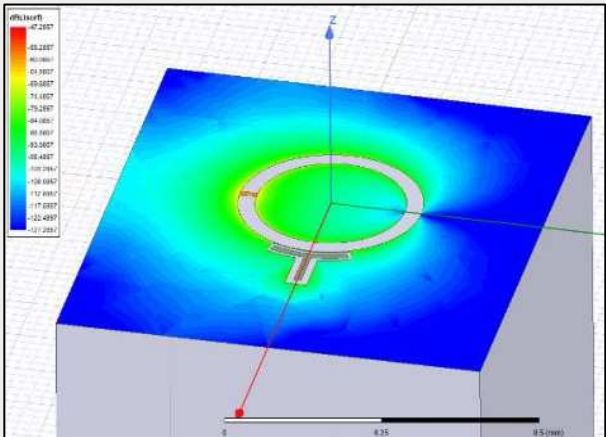
IBM 5Q Tsuru U Tokyo



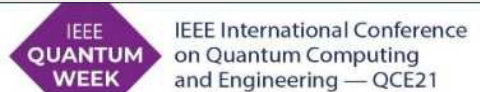
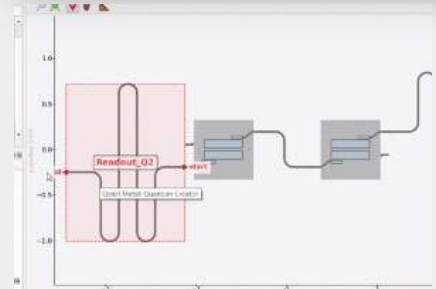
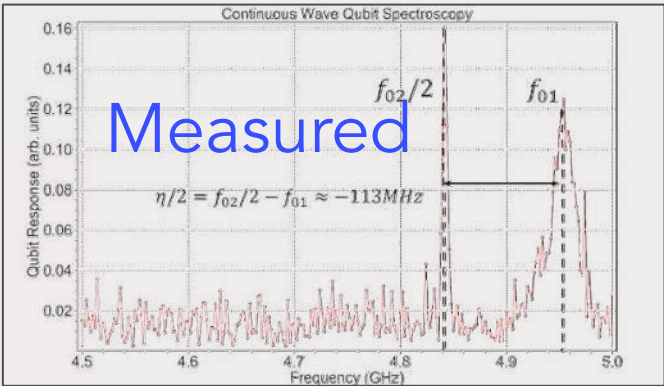
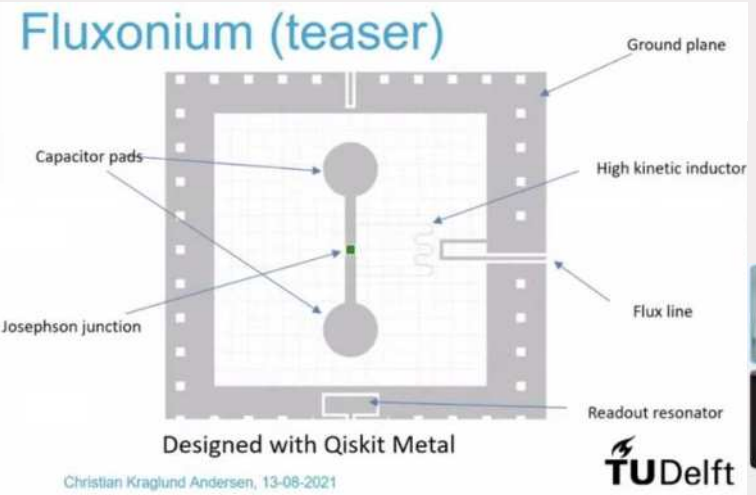
Chalmers 8Q



Simulated



TU Delft Fluxonium
Fluxonium (teaser)



Zlatko Minev, IBM Quantum

The Need for Open Quantum Hardware



Zlatko K. Minev

IBM Quantum

IBM T.J. Watson Research Center
Yorktown Heights, NY



@zlatko_minev



zlatko-minev.com



qiskit.org/metal

Thank you!

“It is enough if one tries merely to comprehend a little of this mystery each day.”

A. Einstein

Ways to learn more



Global Summer Schools QGSS

2020
open access cQED Lecs. 16-21 Minev

2021
Intro to quantum noise



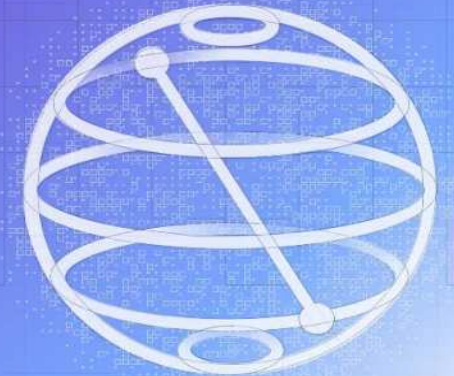
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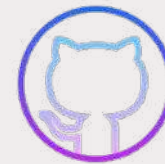


Quantum Seminar Series

(host: Minev; level: advanced)



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access



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Qiskit SDK

An open-source SDK for working with quantum computers at the level of pulses, circuits, and algorithms.

<https://qiskit.org> hello@qiskit.org

Zlatko Minev, IBM Quantum